# In-class exercise

- This week's exercise is to simulate player performance in a soccer/football model.
- The exercise centered on inheritance and polymorphism.
  - *Inheritance*: Building **`Defender`** and **`Striker`** classes from a base **`Player`** class.
  - *Polymorphism*: Overriding base virtual functions to specialize behavior.

- **Base class (`Player`):**
  - Manages the fundamental state, specifically the **`fatigue_level`**.
  - Provides virtual base functions (**`shootingSuccessRate`** and **`tackleSuccessRate`**) that calculate a generic success rate, which factors in the current fatigue.
  - Includes overloaded helper functions (**`checkEnergy`**) for stamina and hydration.

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- **Derived class specialization:**
  - Implement `Defender` and `Striker` derived classes.
  - These classes override the virtual success rate functions to apply a positional multiplier (a bonus or penalty) on top of the base fatigue-adjusted rate (e.g., `Striker` gets a shooting bonus, `Defender` gets a tackling bonus).

- Please begin with `starter.cpp` in LearnUs.
  - There are **Tasks 1 - 12** in the source code.
  - Make sure your complete and run the code without any errors.